Bayes and Empirical Bayes

Contents

1 Beta Distribution

Let us start by recalling the Beta distribution. The Beta distribution with parameters a and b is given by (see, for example, Casella and Berger [2002, Chapter 2]) the density:

$$f_\theta(\theta) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)} I\{0 \leq \theta \leq 1\}$$

where the normalizing constant $B(a,b)$ is the Beta function given by

$$B(a,b) = \int_0^1 \theta^{a-1}(1-\theta)^{b-1} d\theta.$$

For the Beta density to be proper, we need both a and b to be strictly positive. Here are some basic properties of the Beta distribution:

1. When $a = 1$ and $b = 1$, we get the uniform distribution on $(0,1)$. We consider the uniform distribution as the simplest example of a Beta distribution.

2. The mean of the Beta distribution is

$$\text{mean} = \frac{a}{a+b}.$$

   For example, if a is much smaller compared to b, then the mean will be small.

3. The variance of the Beta distribution is

$$\text{variance} = \frac{a}{a+b}\frac{b}{a+b}\frac{1}{a+b+1}.$$

   An implication is that, when $a+b$ is large, the variance tends to be small, so the Beta density will look skinny.

4. More generally, any moment $\mathbb{E}(\theta^k)$ (for positive integers k) can be written in closed form as (here $\theta \sim Beta(a,b)$)

$$\mathbb{E}(\theta^k) = \frac{a(a+1)\dots(a+k-1)}{(a+b)(a+b+1)\dots(a+b+k-1)} \quad \text{for every } k \geq 1.$$

In Bayesian inference, improper priors are also sometimes used. In the case of Beta priors, it is common to use the distributions $Beta(0,0)$ as a prior. This corresponds to the density

$$f_{Beta(0,0)}(\theta) \propto \frac{I\{0 < \theta < 1\}}{\theta(1-\theta)}.$$

This improper density cannot be normalized so we don't put any normalizing constants on the right hand side. If one wants to interpret $Beta(0,0)$ as a probability distribution, the correct answer is the discrete distribution taking the values 0 and 1 with equal probability:

$$Beta(0,0) = Bernoulli(0.5).$$

One way to formalize this equivalence is to argue that the distribution $Beta(\epsilon,\epsilon)$ converges to $Bernoulli(0.5)$ as $\epsilon \downarrow 0$ in the usual sense of weak convergence. Intuitively, the mass around 0 and 1 becomes very large compared to mass in between as ϵ decreases to 0. A rigorous way to see this is that the moments of $Beta(\epsilon,\epsilon)$ converge to the moments of $Bernoulli(0.5)$.

Remark 1 (Visualizing the Beta distribution)**.** *The behavior of the Beta distribution is most easily understood by plotting its density for different values of the parameters a and b. The mean of the distribution is*

$$\frac{a}{a+b},$$

so changing the ratio $a : b$ shifts the center of the density. Meanwhile, the quantity $a+b$ controls the concentration of the distribution around its mean: larger values of $a+b$ produce a more concentrated (less variable) distribution, while smaller values produce a more diffuse distribution. When $a = b = 1$, the Beta distribution is simply the uniform distribution on $(0, 1)$. If $a, b > 1$, the density is unimodal. When both a and b are less than one, the density becomes concentrated near the endpoints 0 and 1. Finally, as $a, b \downarrow 0$, the Beta distribution converges to a two-point distribution assigning equal probability to 0 and 1.

2 Beta-Binomial Inference

The goal is to infer θ from observations X (and n) where $X \sim Bin(n, \theta)$. One can also formulate this problem as that of estimating θ from n i.i.d Bernoulli observations $Z_1, \ldots, Z_n \overset{\text{i.i.d}}{\sim} Ber(\theta)$. The likelihood is given by:

$$f_{X|\theta}(x) = \binom{n}{x}\theta^x(1-\theta)^{n-x} \propto \theta^x(1-\theta)^{n-x}.$$

The frequentist estimator (MLE) for θ is x/n. For Bayesian inference, we will use the $Beta(a, b)$ prior:

$$f_\theta(\theta) \propto \theta^{a-1}(1-\theta)^{b-1} I\{0 \leq \theta \leq 1\}$$

leading to the posterior:

$$\begin{aligned} f_{\theta|x}(\theta) &\propto \theta^{a-1}(1-\theta)^{b-1} I\{0 \leq \theta \leq 1\}\theta^x(1-\theta)^{n-x} \\ &= \theta^{x+a-1}(1-\theta)^{n-x+b-1} I\{0 \leq \theta \leq 1\}. \end{aligned}$$

Thus

$$\theta \mid x \sim Beta(x + a, n - x + b).$$

If $a-1, b-1$ are nonnegative integers then one can interpret the above as if we we have already seen $a - 1$ successes and $b - 1$ failures in independent bernoulli trials.

This Beta-Binomial conjugacy is one of the classical examples of Bayesian inference (see Gelman et al. [2013, Chapter 2]). A natural point estimate of θ is the posterior mean:

$$\hat{\theta} = \mathbb{E}(\theta \mid x) = \frac{x+a}{n+a+b}.$$

Here are two things to note about the posterior mean:

1. It is a convex combination of the MLE and the prior mean $a/(a + b)$:

   $$\frac{x+a}{n+a+b} = \left(\frac{n}{a+b+n}\right)\frac{x}{n} + \left(\frac{a+b}{a+b+n}\right)\frac{a}{a+b}.$$

   So if n is much larger compared to $a + b$, then the posterior mean will be very close to the MLE. Conversely if n is much smaller compared to $a + b$, then the posterior mean will be very close to the prior mean.

2. When $a = b = 0$, then the posterior mean exactly coincides with the MLE. Thus if we use the $Beta(0, 0)$ prior, then posterior inference will be close to frequentist inference. Note again that this prior is improper.

3. If we use the $Beta(0, 0)$ prior and if we observe $n = x = 1$ (i.e., we only have data on one Bernoulli trial which resulted in a heads). Then the posterior is $Beta(x + a, n - x + b) = Beta(1, 0)$. $Beta(1, 0)$ is also improper but it can be interpreted as the point mass at 1:

$$Beta(1, 0) = Bernoulli(1) = \delta_{\{1\}}.$$

This can be proved, for example, by computing the moments of $Beta(1, \epsilon)$ and showing that they all approach 1 (which are the moments of $\delta_{\{1\}}$) as $\epsilon \to 0$. Similarly for $x = 0$ and $n = 1$, the posterior becomes $Beta(0, 1)$ which should be interpreted as $\delta_{\{0\}}$. If $n = 2$ and $x = 1$ (i.e., we observe one head and one tail), the posterior is the uniform distribution on $(0, 1)$.

4. The process of going from the $Beta(0, 0)$ prior to the $Beta(1, 0)$ posterior when $n = x = 1$ is described as intuitive in the following situation by Jaynes [2003, Section 12.4.3]: *...in a chemical laboratory we find a jar containing an unknown and unlabeled compound. We are at first completely ignorant as to whether a small sample of this compound will dissolve in water or not. But, having observed that one small sample does dissolve, we infer immediately that all samples of this compound are water soluble, and although the conclusion does not carry quite the force of deductive proof, we feel strongly that the inference was justified*. I will say before observing the data, we thought θ was either 0 or 1 (we put equal probability on both), so once we see it once we know what it is.

3 Multiple Replications of the Same Problem

Consider the problem of estimating θ_i from (X_i, n_i) for each $i = 1, \ldots, N$, with $X_i \sim Bin(n_i, \theta_i)$. For a concrete example, take all counties in the United States with X_i denoting the number of deaths due to kidney cancer (in the period $1980 - 89$) and n_i denoting the average population (during the same period) for the i-th county. This dataset is discussed extensively by Efron and Hastie [2016, Chapter 1] and we will also look into this in class.

It is easy to see that the naive frequentist estimate X_i/n_i can be very bad for θ_i if n_i is small. We will therefore use Bayes estimation using the prior:

$$\theta_i \overset{\text{i.i.d}}{\sim} Beta(a, b).$$

The choice of a and b is crucial here. Since we have a large dataset, it makes sense to learn good values of a and b from the observed data. We will look at the following three ways of doing this.

1. **Method One**: We place a threshold on n_i and filter out i for which n_i exceeds the threshold. We then select a and b by fitting the $Beta(a, b)$ density to the data $\{X_i/n_i : n_i \geq \text{threshold}\}$. The Beta density fitting can be done by just matching mean and variances. Let m and V denote the mean and variance of $\{X_i/n_i : n_i \geq \text{threshold}\}$. Then we obtain a and b by solving:

$$\frac{a}{a+b} = m \quad \text{and} \quad \frac{a}{a+b}\frac{b}{a+b}\frac{1}{a+b+1} = V$$

which gives

$$\hat{a} = m\left(\frac{m(1-m)}{V} - 1\right) \text{ and } \hat{b} = (1-m)\left(\frac{m(1-m)}{V} - 1\right).$$

One drawback of this method is that the selection of threshold can be arbitrary. In the kidney cancer example, we choose 300000 as the threshold, but we could have also chosen some other threshold such as 350K or 400K.

2. **Method Two**: We consider the marginal likelihood of X_i given a, b. Note that

$$\begin{aligned}
P\{X_i = x \mid a, b\} &= \int_0^1 P\{X_i = x \mid \theta_i = \theta\} f_{\theta_i|a,b}(\theta) d\theta \\
&= \int_0^1 \binom{n_i}{x} \theta^x (1-\theta)^{n_i - x} \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)} d\theta \\
&= \binom{n_i}{x} \frac{1}{B(a,b)} \int_0^1 \theta^{x+a-1}(1-\theta)^{n_i-x+b-1} d\theta \\
&= \binom{n_i}{x} \frac{B(x+a, n_i - x + b)}{B(a,b)}
\end{aligned}$$

As a result, the marginal likelihood of the data given in terms of a, b is:

$$\prod_{i=1}^{N} \binom{n_i}{X_i} \frac{B(X_i + a, n_i - X_i + b)}{B(a,b)}$$

So the maximum likelihood estimates of a, b are given by:

$$(\hat{a}_{\text{MLE}}, \hat{b}_{\text{MLE}}) = \underset{a,b}{\text{argmax}} \prod_{i=1}^{N} \binom{n_i}{X_i} \frac{B(X_i + a, n_i - X_i + b)}{B(a,b)}.$$

This maximization can be done numerically by taking a grid of a and b values.

3. **Method Three**: This is the fully Bayes solution where we simply place priors on a and b. It is sensible to reparametrize as:

$$\mu = \frac{a}{a+b} \quad \text{and} \quad \kappa = a + b.$$

Now we take the priors:

$$\mu \sim \text{uniform}(0,1) \quad \text{and} \quad \log \kappa \sim \text{uniform}(-\infty, \infty).$$

Inference under this fully Bayesian model is nontrivial from a computational point of view. We shall discuss this in class.

For the kidney cancer dataset $(X_i, n_i), i = 1, \ldots, N$ (N denotes the total number of counties, n_i is the average population of county i and X_i is the number of deaths due to kidney cancer in the fixed time period $1980 - 89$). Consider the following three models for this dataset:

1. **Model One**: $\theta_i \overset{\text{i.i.d}}{\sim} Beta(0,0)$ and $X_i \mid \theta_i \overset{\text{ind}}{\sim} Bin(n_i, \theta_i)$. This model uses an uniformative prior on θ_i, and the posterior mean estimate of each θ_i coincides with the frequentist MLE X_i/n_i. This model performs poorly for the present dataset. In particular, rankings of counties by estimated risk (e.g., the top 100 values of $\hat{\theta}_i$) are dominated by counties with small populations. Moreover, the model yields inaccurate predictions for future county-level death counts.

2. **Model Two**: $\theta_i \overset{\text{i.i.d}}{\sim} Beta(15, 150000)$ and $X_i \mid \theta_i \overset{\text{ind}}{\sim} Bin(n_i, \theta_i)$. This model employs a highly informative prior on θ_i, where the hyperparameters $(15, 150000)$ may be viewed as arising from a Beta distribution fitted to historical data. The prior encodes strong prior knowledge about the prevalence of kidney cancer mortality. This leads to substantial shrinkage of the posterior estimates towards the prior mean. Unsurprisingly, given the strength of the prior information, this model yields good empirical performance on the dataset.

3. **Model Three**: $\log a, \log b \overset{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty)$, $\theta_i \overset{\text{i.i.d}}{\sim} Beta(a, b)$, $X_i \mid \theta_i \overset{\text{ind}}{\sim} Bin(n_i, \theta_i)$. This model places an uninformative prior on the hyperparameters a and b, and hence does not inject strong prior knowledge about the prevalence of kidney cancer mortality. Compared to Models One and Two, this is a substantially more flexible and principled model, as it allows the amount of shrinkage to be learned from the data rather than fixed a priori.

   When fitted to the data, the posterior distribution of (a, b) concentrates sharply around values close to $(15, 150000)$ (see the code file for this lecture), indicating that the model successfully infers from the data that kidney cancer deaths are rare events. As a consequence, the posterior estimates of the individual θ_i are adaptively shrunk toward a common mean, with the degree of shrinkage depending on the corresponding population sizes n_i. This hierarchical borrowing of strength leads to stable estimation, sensible county-level rankings, and accurate predictions of future death counts, and overall the model performs very well.

   Note that this model does not require any a priori knowledge specific to kidney cancer and is therefore broadly applicable. For example, the same hierarchical framework can be used to predict batting averages in the baseball example, which will be explored in a homework question.

4 Bayesian Inference with Normal Likelihoods

4.1 Normal Approximation to Binomial and Variance Stabilizing Transformation

We will next discuss Bayesian inference with normal likelihoods, and also study some connections to the James-Stein estimator. Normal likelihoods are applicable even in Binomial situations, such as in the baseball data analysis example from Efron and Hastie [2016, Chapter 1]. Here X_i denotes the number of hits in $n_i = n = 45$ at bats for player i. The natural model is: $X_i \sim Bin(n_i, p_i)$ with parameter p_i but a normal likelihood (as opposed to Binomial likelihood) can also be used because by invoking the Central Limit Theorem, we can write

$$\frac{X_i}{n_i} \overset{\text{approx}}{\sim} N\left(p_i, \frac{p_i(1-p_i)}{n_i}\right).$$

This is because:

$$\sqrt{n}\left(\frac{X_i}{n_i} - p_i\right) \overset{\text{Law}}{\to} N(0, p_i(1-p_i)) \text{ as } n \to \infty. \tag{1}$$

Note that the unknown parameter p_i appears in the variance above. If we want to work with normal likelihoods with known variance, a clean way is to use a variance stabilizing transformation. This is justified by the use of Delta method.

Theorem 1 (Delta Method)**.** *(see, e.g., Casella and Berger [2002, Chapter 5]) If* $\sqrt{n}(T_n - p) \overset{Law}{\to} N(0, \tau^2)$ *as* $n \to \infty$*, then*

$$\sqrt{n}\left(g(T_n) - g(p)\right) \overset{Law}{\to} N(0, \tau^2 (g'(p))^2)$$

as $n \to \infty$ *provided* $g'(p)$ *exists and is non-zero.*

Informally, the Delta method states that if T_n has a limiting Normal distribution, then $g(T_n)$ also has a limiting normal distribution and also gives an explicit formula for the asymptotic variance of $g(T_n)$. This is surprising because g can be linear or non-linear. In general, non-linear functions of normal random variables do not have a normal distribution. But the Delta method works because under the assumption that $\sqrt{n}(T_n - p) \overset{\text{Law}}{\to} N(0, \tau^2)$, it follows that T_n converges in probability to p so that T_n will be close to p at least for large n. In a neighborhood of p, the non-linear function g can be approximated by a linear function which means that g effectively behaves like a linear function. Indeed, the Delta method is a consequence of the approximation:

$$g(T_n) - g(p) \approx g'(p)\,(T_n - p)\,.$$

By the Delta method and (1), we have

$$\sqrt{n_i}\left(g(X_i/n_i) - g(p_i)\right) \overset{\text{Law}}{\to} N(0, (g'(p_i))^2 p_i(1-p_i)).$$

The variance above will not depend on p_i if we choose the function g so that

$$g'(p_i) = \frac{1}{\sqrt{p_i(1-p_i)}}$$

Solving this for g, we get $g(p_i) = 2\arcsin(\sqrt{p_i})$. This is a variance stabilizing transformation for the Binomial. Thus by the Delta method, we have

$$2\sqrt{n}\left(\arcsin(\sqrt{X_i/n_i}) - \arcsin(\sqrt{p_i})\right) \overset{\text{Law}}{\to} N(0,1).$$

While working with binomial counts, it thus makes sense to transform the observation X_i and parameter p_i into

$$Y_i := 2\sqrt{n_i}\arcsin(\sqrt{X_i/n_i}) \quad \text{and} \quad \theta_i := 2\sqrt{n_i}\arcsin(\sqrt{p_i})$$

respectively and then work with the likelihood

$$Y_i|\theta_i \sim N(\theta_i, 1).$$

4.2 Normal Normal Conjugacy

We will now study Bayesian estimation under the model where data is y and parameter is θ, the likelihood is given by:

$$y \mid \theta \sim N(\theta, \sigma^2)$$

for a fixed $\sigma > 0$. We will look at the case of unknown σ later. The frequentist estimator of θ is the MLE y. For Bayesian inference, we use the normal $N(\mu, \tau^2)$ prior on θ. The basic fact is:

$$\theta \sim N(\mu, \tau^2) \;\text{ and }\; y \mid \theta \sim N(\theta, \sigma^2) \implies \theta \mid y \sim N\left(\frac{y/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2}, \frac{1}{1/\sigma^2 + 1/\tau^2}\right)$$
$$\text{and } y \sim N(\mu, \sigma^2 + \tau^2).$$

The mean of the posterior distribution is thus

$$\frac{y/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2} = \frac{1/\sigma^2}{1/\sigma^2 + 1/\tau^2} y + \frac{1/\tau^2}{1/\sigma^2 + 1/\tau^2} \mu$$

which is a weighted linear combination of the prior mean μ and the data y. The weights are inversely proportion to the variances of the corresponding normal distributions. For a normal distribution, we use the term "precision" to denote the inverse of variance. Thus the weights of the linear combination above are proportional to the precisions of the prior and the likelihood.

Also note that the precision of the posterior equals the sum of the prior and likelihood precisions.

Note that when $\tau^2 \to \infty$ and μ is a fixed constant, we have $\theta \mid y \sim N(y, \sigma^2)$ (in this case, the posterior mean coincides with the MLE y). Operationally, the $N(\mu, \tau^2)$ prior with fixed μ and $\tau^2 = +\infty$ has the same behavior as the uniform$(-\infty, \infty)$ prior. These are uninformative priors in this problem.

Often the data will consist of multiple numbers $y_1, \ldots, y_n$ with the likelihood being:

$$y_1, \ldots, y_n \overset{\text{i.i.d}}{\sim} N(\theta, \sigma^2).$$

In this case (for the same prior $\theta \sim N(\mu, \tau^2)$), the posterior is

$$\theta | y_1, \ldots, y_n \sim N\left(\frac{n\bar{y}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2}, \frac{1}{n/\sigma^2 + 1/\tau^2}\right).$$

where $\bar{y} := (y_1 + \cdots + y_n)/n$. This can be proved directly or use the fact that $\bar{y}$ is sufficient for θ and the data can be reduced to the single number $\bar{y}$ with likelihood $N(\theta, \sigma^2/n)$.

5 Multiple Instances of the Same Problem

Consider the problem of estimating $\theta_1, \ldots, \theta_N$ from data $y_1, \ldots, y_N$ under the likelihood:

$$y_i \overset{\text{ind}}{\sim} N(\theta_i, \sigma^2)$$

for a fixed σ^2. We further assume:

$$\theta_i \overset{\text{i.i.d}}{\sim} N(\mu, \tau^2).$$

If we fix μ to be some constant value (say $\mu = 0$) and τ^2 to be very large, then the estimate of θ_i would be equal to y_i.

Instead of fixing these values, we can treat μ and τ^2 also as unknown parameters and attempt to estimate them from the observed data. Marginalizing θ_i, it is easy to see that

$$y_i \mid \mu, \tau \overset{\text{i.i.d}}{\sim} N(\mu, \sigma^2 + \tau^2).$$

One can estimate μ and τ from this model by some estimates $\hat{\mu}$ and $\hat{\tau}$. θ_i is then estimated by $\mathbb{E}(\theta_i \mid y_i, \mu = \hat{\mu}, \tau = \hat{\tau})$. How to obtain $\hat{\mu}$ and $\hat{\tau}$? It is natural to take

$$\hat{\mu} = \frac{y_1 + \cdots + y_N}{N}.$$

The estimate of τ should be based on the sample variance:

$$\sum_{i=1}^{N} (y_i - \bar{y})^2 \sim \left(\sigma^2 + \tau^2\right) \chi^2_{N-1}.$$

The formula for $\mathbb{E}(\theta_i \mid y_i, \mu, \tau)$ is:

$$\mathbb{E}(\theta_i \mid y_i, \mu, \tau) = \frac{y_i/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2} = \mu + \left(1 - \frac{\sigma^2}{\sigma^2 + \tau^2}\right)(y_i - \mu).$$

So we need to estimate $1/(\sigma^2 + \tau^2)$ as opposed to τ^2 directly. The following fact:

$$\mathbb{E}\left(\frac{1}{\chi^2_v}\right) = \frac{1}{v-2}, \qquad v > 2.$$

shows that

$$\mathbb{E}\left(\frac{1}{\sum_{i=1}^n (y_i - \bar{y})^2}\right) = \frac{1}{\sigma^2 + \tau^2}\mathbb{E}\left(\frac{1}{\chi^2_{N-1}}\right) = \frac{1}{(N-3)(\sigma^2 + \tau^2)}.$$

So we use

$$\text{estimate of } \frac{1}{\sigma^2 + \tau^2} = \frac{N-3}{\sum_{i=1}^n (y_i - \bar{y})^2}. \tag{2}$$

The estimate of τ^2 implied by (2) can be nonpositive (depending on the data and σ^2). We shall ignore this however.

The estimates of θ_i are therefore given by:

$$\hat{\theta}_i^{\text{JS}} := \bar{y} + \left(1 - \frac{(N-3)\sigma^2}{\sum_{i=1}^n (y_i - \bar{y})^2}\right)(y_i - \bar{y}).$$

This is the James-Stein estimator. The estimator was introduced by James and Stein [1961] following the celebrated inadmissibility result of Stein [1956].

The James-Stein estimator can thus be seen as an **empirical Bayes** procedure. For a modern discussion of empirical Bayes methods, see Efron and Hastie [2016, Chapters 1–3]. The James-Stein estimator has the following remarkable frequentist property: its risk is unformly better than the naive estimator $\hat{\theta}_{i,\text{naive}} = y_i$ in mean squared error for all values of $\theta_1, \ldots, \theta_N$ (this is true for $N \geq 3$; although we only defined it for $N > 3$, one can create a more naive version of James-Stein replacing $\bar{y}$ by any fixed constant like 0 and this would work also for $N = 3$). This property is beyond the scope of this class.

Let us now present the full Bayes estimate. This requires specification of (uninformative) priors of μ and τ. For μ, we use the uniform$(-\infty, \infty)$ prior. For τ, we use the fact that the marginal distribution of the data given μ, τ is $y_i \mid \mu, \tau \overset{\text{i.i.d}}{\sim} N(\mu, \sigma^2 + \tau^2)$, which is in terms of the parameter $\gamma^2 := \sigma^2 + \tau^2$. For the $N(\mu, \gamma^2)$ model, the standard uninformative prior for γ is

$$\log \gamma \sim \text{uniform}(-\infty, \infty) \quad \text{or} \quad f_\gamma(\gamma) \propto \frac{I\{\gamma > 0\}}{\gamma}.$$

Now we have the additional information that $\gamma > \sigma$ (because $\gamma^2 = \sigma^2 + \tau^2$), so it is natural to modify the above prior as:

$$f_\gamma(\gamma) \propto \frac{I\{\gamma > \sigma\}}{\gamma}.$$

It is easy to check that this prior on γ leads to the following prior on $\tau = \sqrt{\gamma^2 - \sigma^2}$:

$$f_\tau(\tau) \propto \frac{\tau}{\sigma^2 + \tau^2} I\{\tau > 0\}.$$

To recap, we are using the following prior for μ and τ:

$$\mu \sim \text{uniform}(-\infty, \infty) \quad \text{and} \quad f_\tau(\tau) \propto \frac{\tau}{\sigma^2 + \tau^2} I\{\tau > 0\}.$$

We will combine this with the likelihood $y_i \mid \mu, \tau \overset{\text{i.i.d}}{\sim} N(\mu, \sigma^2 + \tau^2)$ to obtain the posterior of μ, τ. It can be checked that:

$$\mu \mid \gamma, y_1, \ldots, y_N \sim N\left(\bar{y}, \gamma^2/N\right) \text{ and } f_{\gamma|y_1,\ldots,y_N}(\gamma) \propto \gamma^{-N} \exp\left(-\frac{\sum_{i=1}^N (y_i - \bar{y})^2}{2\gamma^2}\right) I\{\gamma > \sigma\}.$$

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